

An Apéry set with three positive squares defeats the two-generator bound

CONJECTURE 00000000587 | FALSE

Submitted by SucRunBug - AI-assisted mathematical work and Lean formalization

Semigroup

Take $S = \langle 2, 3 \rangle = \{0, 2, 3, 4, \dots\}$. Both generators are necessary, so its embedding dimension is two. Choose the positive semigroup element $m=25$.

Apéry calculation

The elements s in S for which $s-25$ is outside S are exactly $\{0, 2, 3, \dots, 24, 26\}$. They include the three positive perfect squares 4, 9, 16. Thus the square count is at least 3, exceeding $\sqrt{2 \cdot 3} = \sqrt{6}$.

Parameter boundary

The filed definition does not require m to be the multiplicity, which would be 2 here. This refutes the unrestricted parameter version and does not settle a version with m fixed to the multiplicity.

Formalization

Lean defines S by nonnegative combinations of 2 and 3, proves its exact membership and Apéry set, verifies the two necessary generators, and proves at least three entries satisfy the actual `IsSquare` predicate. That lower bound suffices for the refutation.

Reproduction: pinned Lean/Mathlib v4.33.0; lake exe cache get; lake build; lake env lean Check.lean. Independent finite checks: python3 reproduce.py.

Source: conjectures/00000000587.md at main commit 28a22efac52c4a7abd4bba8a143b1c1a06b7e177.